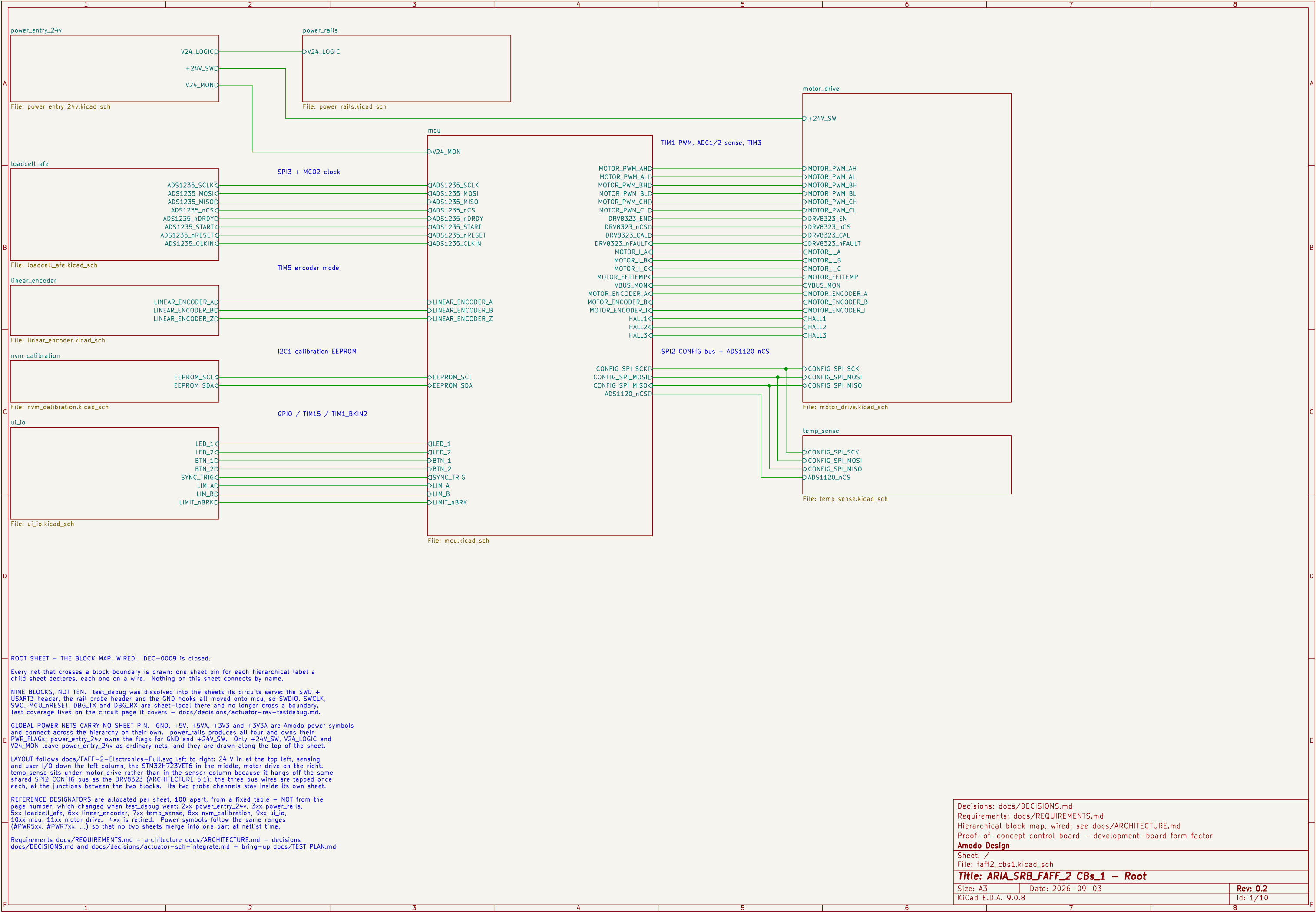




power_rails

V24_LOGIC

File: power_rails.kicad_sch

mcu

V24_MON

SPI3 + MC02 clock

TIM5 encoder mode

I2C1 calibration EEPROM

GPIO / TIM15 / TIM1_BKIN2

File: mcu.kicad_sch

TIM1 PWM, ADC1/2 sense, TIM3

MOTOR_PWM_AHD

MOTOR_PWM_ALD

MOTOR_PWM_BHD

MOTOR_PWM_BLD

MOTOR_PWM_CHD

MOTOR_PWM_CLD

DRV8323_END

DRV8323_nCSD

DRV8323_CALD

DRV8323_nFAULT

MOTOR_I_A

MOTOR_I_B

MOTOR_I_C

MOTOR_FETTEMP

VBUS_MON

MOTOR_ENCODER_A

MOTOR_ENCODER_B

MOTOR_ENCODER_I

HALL1

HALL2

HALL3

SPI2 CONFIG bus + ADS1120 nCS

CONFIG_SPI_SCKD

CONFIG_SPI_MOSID

CONFIG_SPI_MISO

ADS1120_nCSD

File: mcu.kicad_sch

motor_drive

+24V_SW

MOTOR_PWM_AH

MOTOR_PWM_AL

MOTOR_PWM_BH

MOTOR_PWM_BL

MOTOR_PWM_CH

MOTOR_PWM_CL

DRV8323_EN

DRV8323_nCS

DRV8323_CAL

DRV8323_nFAULT

MOTOR_I_A

MOTOR_I_B

MOTOR_I_C

MOTOR_FETTEMP

VBUS_MON

MOTOR_ENCODER_A

MOTOR_ENCODER_B

MOTOR_ENCODER_I

QHALL1

QHALL2

QHALL3

CONFIG_SPI_SCK

CONFIG_SPI_MOSI

CONFIG_SPI_MISO

File: motor_drive.kicad_sch

temp_sense

CONFIG_SPI_SCK

CONFIG_SPI_MOSI

CONFIG_SPI_MISO

ADS1120_nCS

File: temp_sense.kicad_sch

ROOT SHEET – THE BLOCK MAP, WIRED. DEC–0009 is closed.

Every net that crosses a block boundary is drawn: one sheet pin for each hierarchical label a child sheet declares, each one on a wire. Nothing on this sheet connects by name.

NINE BLOCKS, NOT TEN. test_debug was dissolved into the sheets its circuits serve: the SWD + USART3 header, the rail probe header and the GND hooks all moved onto mcu, so SWDIO, SWCLK, SWO, MCU_nRESET, DBG_TX and DBG_RX are sheet–local there and no longer cross a boundary. Test coverage lives on the circuit page it covers – docs/decisions/actuator–rev–testdebug.md.

GLOBAL POWER NETS CARRY NO SHEET PIN. GND, +5V, +5VA, +3V3 and +3V3A are Amodo power symbols and connect across the hierarchy on their own. power_rails produces all four and owns their PWR_FLAGS: power_entry_24v owns the flags for GND and +24V_SW. Only +24V_SW, V24_LOGIC and V24_MON leave power_entry_24v as ordinary nets, and they are drawn along the top of the sheet.

LAYOUT follows docs/FAFF–2–Electronics–Full.svg left to right: 24 V in at the top left, sensing and user I/O down the left column, the STM32H723VET6 in the middle, motor drive on the right, temp_sense sits under motor_drive rather than in the sensor column because it hangs off the same shared SPI2 CONFIG bus as the DRV8323 (ARCHITECTURE 5.1); the three bus wires are tapped once each, at the junctions between the two blocks. Its two probe channels stay inside its own sheet.

REFERENCE DESIGNATORS are allocated per sheet, 100 apart, from a fixed table – NOT from the page number, which changed when test_debug went: 2xx power_entry_24v, 3xx power_rails, 5xx loadcellLafe, 6xx linear_encoder, 7xx temp_sense, 8xx nvm_calibration, 9xx ui_io, 10xx mcu, 11xx motor_drive, 4xx is retired. Power symbols follow the same ranges (#PWR5xx, #PWR7xx, ...) so that no two sheets merge into one part at netlist time.

Requirements docs/REQUIREMENTS.md – architecture docs/ARCHITECTURE.md – decisions docs/DECISIONS.md and docs/decisions/actuator–sch–integrate.md – bring–up docs/TEST_PLAN.md

Decisions: docs/DECISIONS.md

Requirements: docs/REQUIREMENTS.md

Hierarchical block map, wired; see docs/ARCHITECTURE.md

Proof–of–concept control board – development–board form factor

Amodo Design

Sheet: /

File: faff2_cbs1.kicad_sch

Title: ARIA_SRB_FAFF_2 CBs_1 – Root

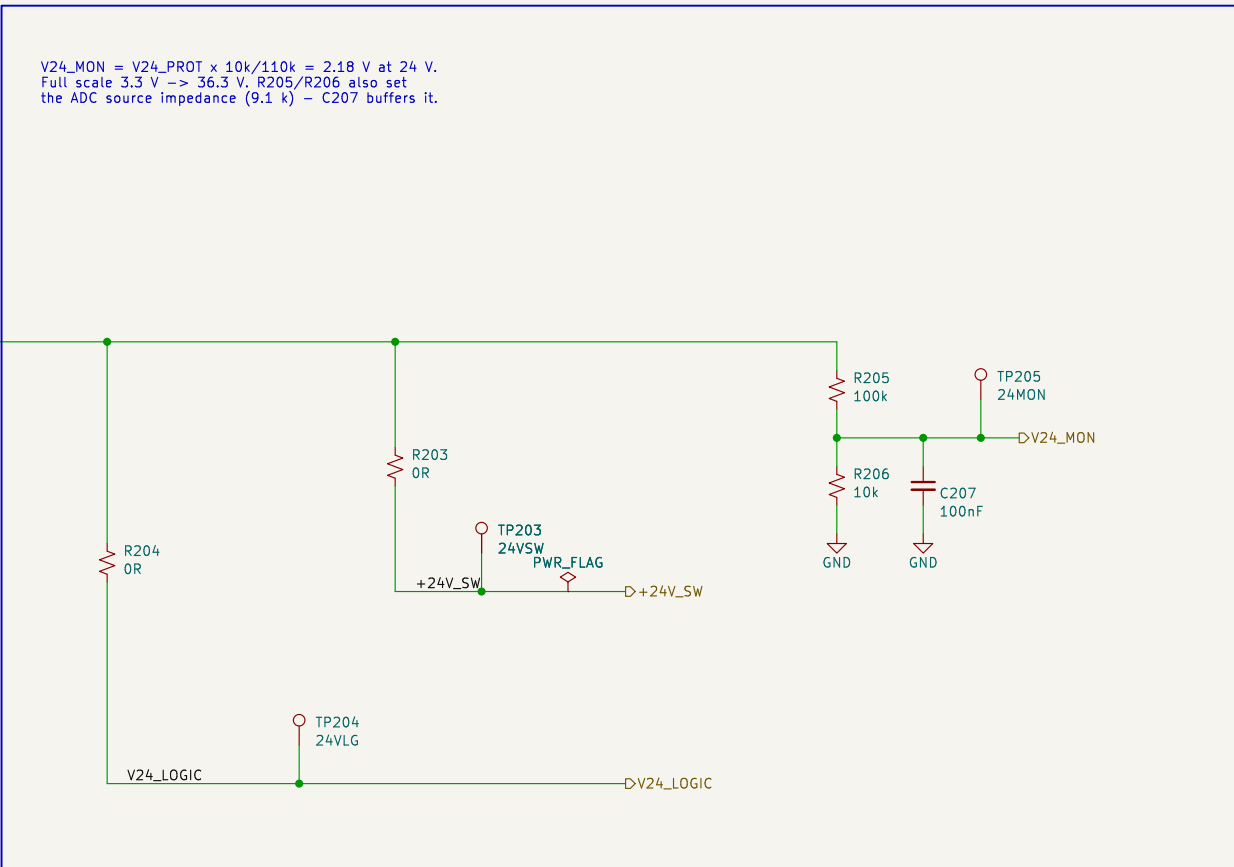
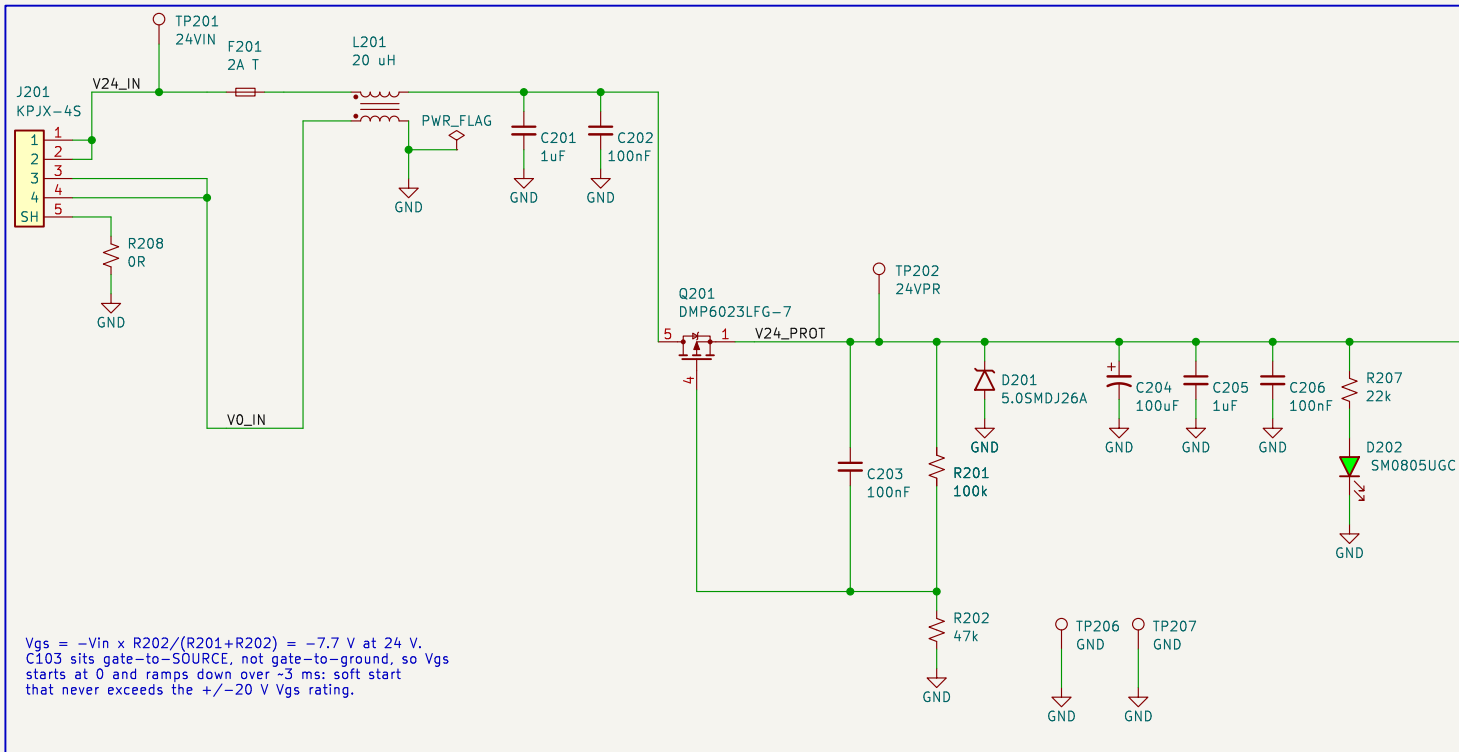
Size: A3

Date: 2026–09–03

Rev: 0.2

KiCad E.D.A. 9.0.8

Id: 1/10



BLOCK: power_entry_24v – 24 V DC input and protection (REQ-EL-01..04)

```
CHAIN J201.KPJX-45[contacts doubled: 1,2 = +24 V, 3,4 = 0 V, 5 = shell]
-> F201 2 A time-lag (25 W peak = 1.04 A at 24 V; 2 A I holds 1.5 A)
-> L201 common-mode choke, both lines; board GND is defined on its load side
-> Q201 P-channel reverse-polarity series switch (drain = supply, source = load,
so the body diode blocks a reversed supply and nothing downstream conducts)
-> D201 TVS + C204..C206 bulk/HF -> V24_PROT, the protected 24 V bus.
```

BRANCHES (docs/TES4_PLAN.md 3.3 – each an isolation link AND a current break)
R203 -> +24V_SW : motor_drive only. Open R203 and the whole board can be exercised with no possibility of the actuator moving.
R204 -> V24_LOGIC: power_rails only. Independent of R203.
One OR link per branch serves as both the isolation link and the current break; a second series part would add joints and no capability.

GROUND One GND net board-wide. AGND / DGND / PGND in the block stub notes are layout partitions of it, not separate nets – see the decisions file.
This sheet owns the single GND PWR_FLAG; no other sheet may add one.

```

INTERFACES exported (hierarchical labels, wired to root sheet pins - DEC-0009)
  OUT +24V_SW  -> motor_drive          OUT V24_LOGIC -> power_rails
  OUT V24_MON  -> mcu PC1 / ADC1_INP11
  Global power nets used here: GND.

```

Reasoning, part sizing and residual ERC: docs/decisions/actuator-sch-power.md

Design reasoning: docs/decisions/actuator-sch-power.md
Requirements: REQ-EL-01..04 – docs/REQUIREMENTS.md
Independent motor / logic branch links (TEST_PLAN 3.3)
24 V in on KPJX-4S: fuse, CM choke, reverse polarity, TVS

Amodo Design

Sheet: /power_entry_24v/
File: power_entry_24v.kicad_sch

Title: CBs_1 – Power Entry 24 V

Size: A3	Date: 2026-09-02	Rev: 0.2
KiCad E.D.A. 9.0.8		Id: 2/10

A. PRE-REGULATOR V24_LOGIC -> 6.0 V (LMR33630, 400 kHz)

B. POST-REGULATED 5 V RAILS (ADPL42005: +5V digital, +5VA analog)

6.0 V, not 5.0 V: it is the ADPL42005's own characterised VIN = VOUT + 1 V point, so the LDOs meet their noise and PSRR numbers on +5VA, which carries REQ-FF-04. R303/R304 are sized so the worst-case floor clears 6.0 V, not just the nominal: 6.009 / 6.110 / 6.212 V. DEC-P10 has the stack.

EN is tied to each LDO input – the datasheet's own automatic-startup connection – so the 5 V rails follow the pre-regulator with no sequencing logic. SENSE ties to VOUT at the regulator, not past the OR link, so lifting the link cannot open the loop. PG is unused: no-connect, not tied to RAIL_PG00D. +5VA feeds ADS1235 / ADS1120 AVDD and the AFE excitation chain; +5V feeds the IKP11 read head.

B. POST-REGULATED 5 V RAILS (ADPL42005: +5V digital, +5VA analog)

EN is tied to each LDO input – the datasheet's own automatic-startup connection – so the 5 V rails follow the pre-regulator with no sequencing logic. SENSE ties to VOUT at the regulator, not past the OR link, so lifting the link cannot open the loop. PG is unused: no-connect, not tied to RAIL_PGGOOD. +5VA feeds ADS1235 / ADS1120 AVDD and the AFE excitation chain; +5V feeds the IKP11 read head.

The schematic diagram illustrates the LMR33630 buck converter circuit. The central component is the U304 LMR33630 IC, which is configured with the following components:

- Input Stage:** The VIN pin (pin 2) is connected to the V24_LOG1ED input through a 100k resistor (R308). The EN pin (pin 3) is connected to GND through an 8.2k resistor (R309). The PGND pin (pin 1) is connected to GND through a 1uF capacitor (C318).
- Output Stage:** The SW pin (pin 8) is connected to the output node through a 100nF capacitor (C319). The FB pin (pin 5) is connected to the output node through a 43k resistor (R311). The RAIL_PG00D pin (pin 4) is connected to the output node through a 1k resistor (R316).
- Output Filter:** The output node is connected to the +3V3 output through a 6.8uH inductor (L302). The output filter is composed of three capacitors: a 22uF capacitor (C320), a 22uF capacitor (C321), and a 100nF capacitor (C322).
- Other Components:** The PWR_FLAG pin (pin 7) is connected to the output node through a 0R resistor (R312). The output node is also connected to the +3V3 output through a 0R resistor (R312).

The circuit is powered by a +3V3 input and produces a +3V3 output. The output is connected to the TP306 test point.

PH1-06-UA
J301

1 2 3 4 5 6

+5V +3V3 +3V3A

GND

FB301
600R@100MHz

+3V3

R313
100k

C323
10uF

C324
100nF

GND

+3V3A

PWR_FLAG

+3V3

R314
330R

D303
SM0805URC

TP307
+3V3A

TP308
PGOOD

RAIL_PG00D

TP311
GND

TP312
GND

GND

GND

RAIL PROBE HEADER (TEST_PLAN 4)
One header to check every logic rail with a DMM during bring-up steps 2 and 3 of TEST_PLAN 5. It only observes: the rails, their isolation links and their current breaks are the rest of this sheet.
1 +5V 2 +3V3 3 +3V3A 4 GND 5 GND 6 GND

+24V IS DELIBERATELY NOT ON THIS HEADER. 24 V next to 3.3 V on a 2.54 mm strip is one slip away from destroying the board; power_entry_24v carries the +24V_SW test point and the current break that TEST_PLAN 4 asks for.

Moved here from mcu in review round 4. Its old note named this sheet as the alternative home; the captain chose it.

FB301 is the +3V3A isolation link and current break: lift it and the ADC digital supplies are cut loose.

POWER SOURCE DECLARATIONS
One PWR_FLAG per rail produced here, at the regulator that makes it, on the load side of that rail's OR link. No other sheet may add one for these nets.

- RAILS +6V0 LMR33630 buck from V24.LOGIC, 6.110 V nominal. Pre-regulator only, local to this sheet. DEC-1P10.
- +5V ADPL42005 LDO from +6V0. IKP11 read head (<65 mA plus RS-422 drive) and spare.
- +5VA ADPL42005 LDO from +6V0. ADS1235 / ADS1200 AVDD and the AFE 5 V excitation and reference chain, which lives in loadcellAFE. Take it from +5VA, not +5V.
- +3V3 LMR33630 buck from V24.LOGIC. STM32, USB3320, DRV8323 logic, QSPI, EEPROM, RS-422 receiver, rotary encoder. -1.1 A worst case against a 3 A part.
- +3V3A ferrite-isolated from +3V3 for the ADC digital supplies.

WHY A PRE-REGULATOR The analog 5 V must be regulated separately from the digital 5 V or the read head switching current lands on the ADC supply, and an LDO needs headroom above 5 V to do it. One buck at 6.0 V feeding two LDOs costs one converter and post-regulates both 5 V rails, 400 kHz on both bucks; 3.3 V at 1.4 MHz would need a -98 ns on-time, too close to the minimum.

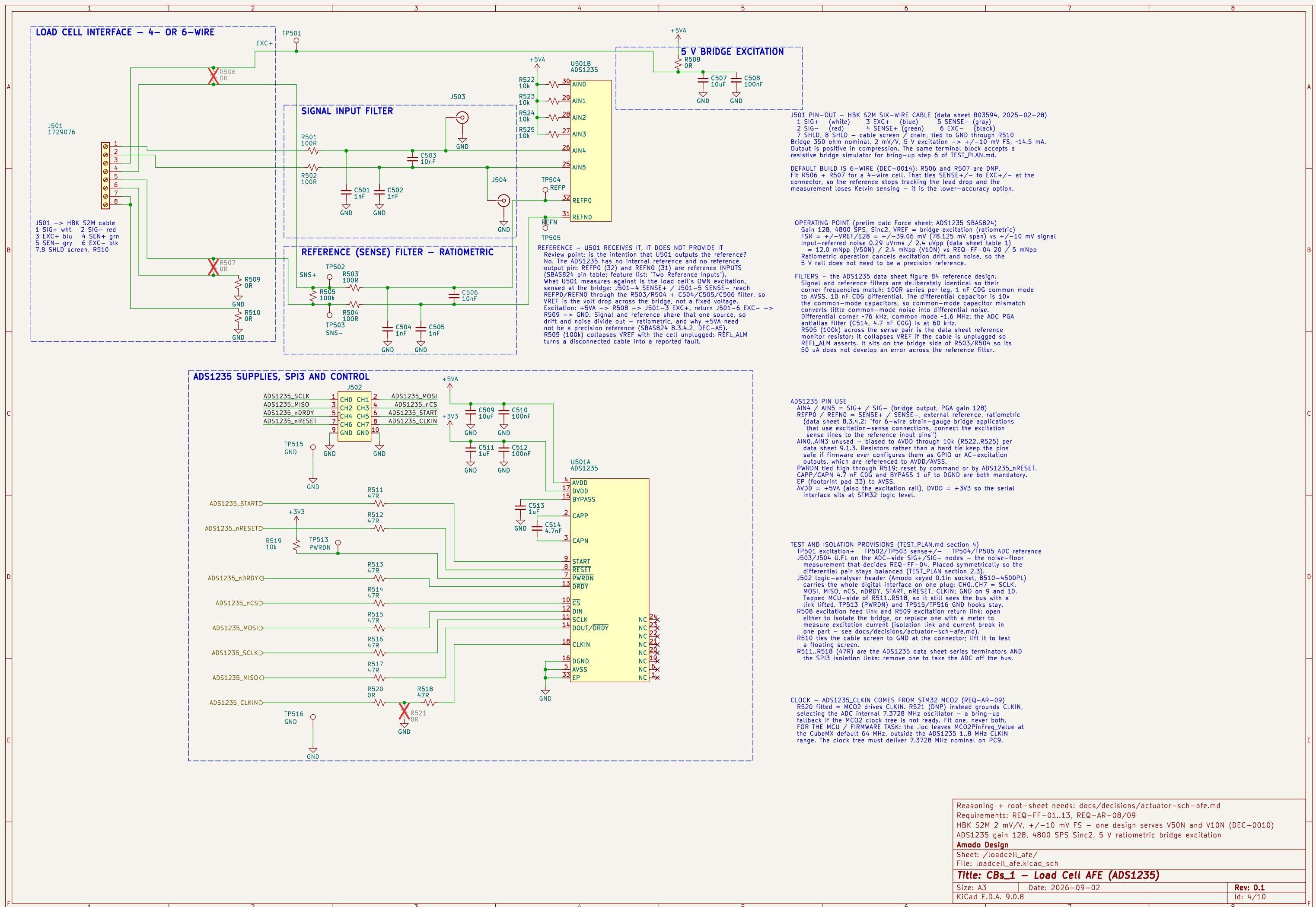
SEQUENCING None required. The shared EN divider (16.2 V rising) holds both converters off until the 24 V input is real; the LDOs follow +6V0 and +3V3A follows +3V3.

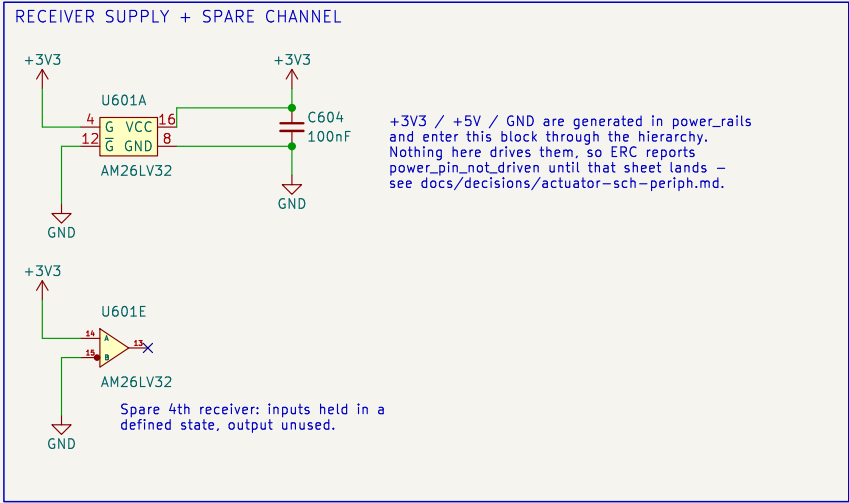
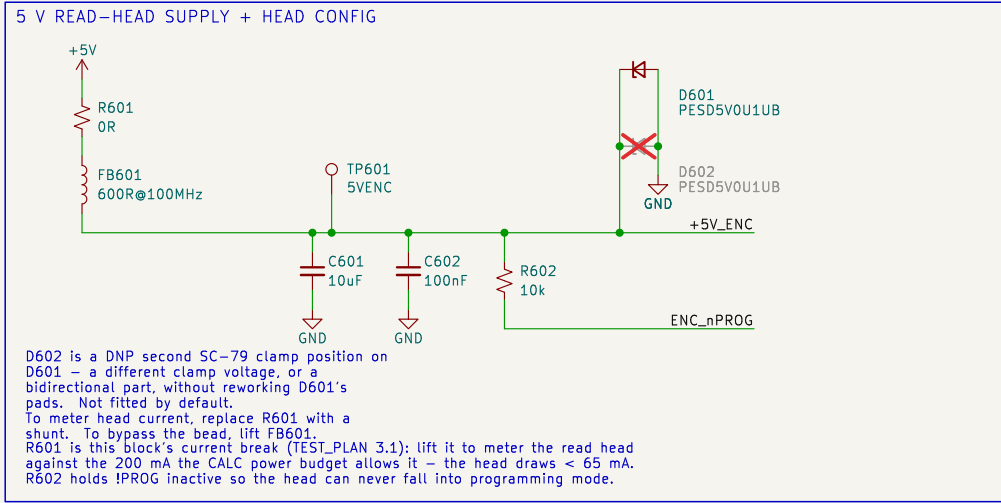
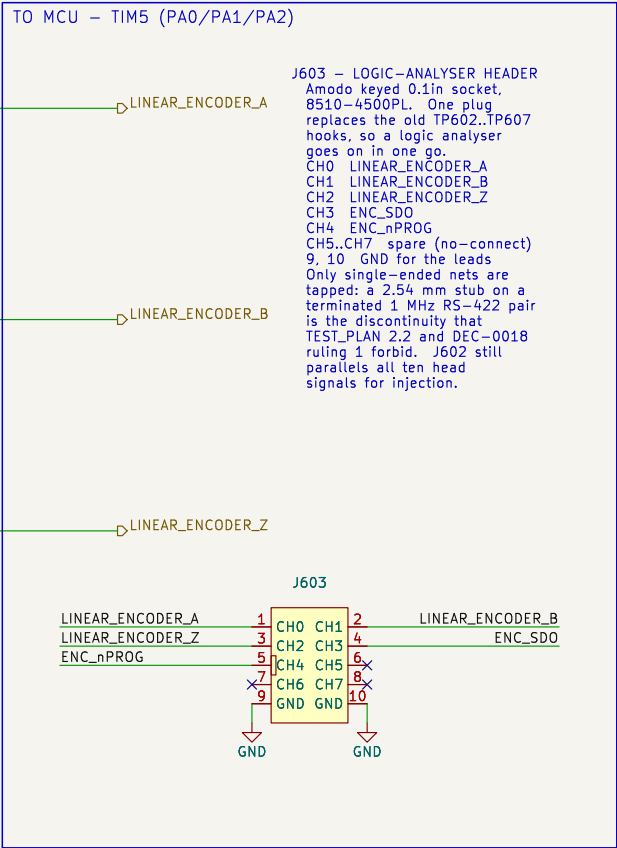
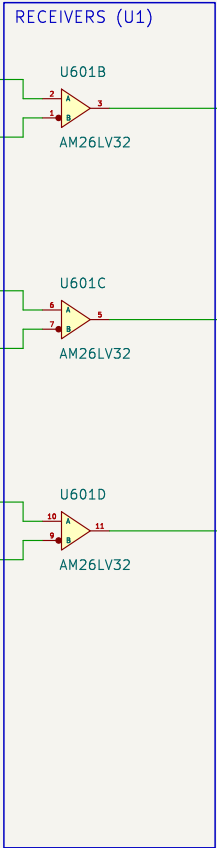
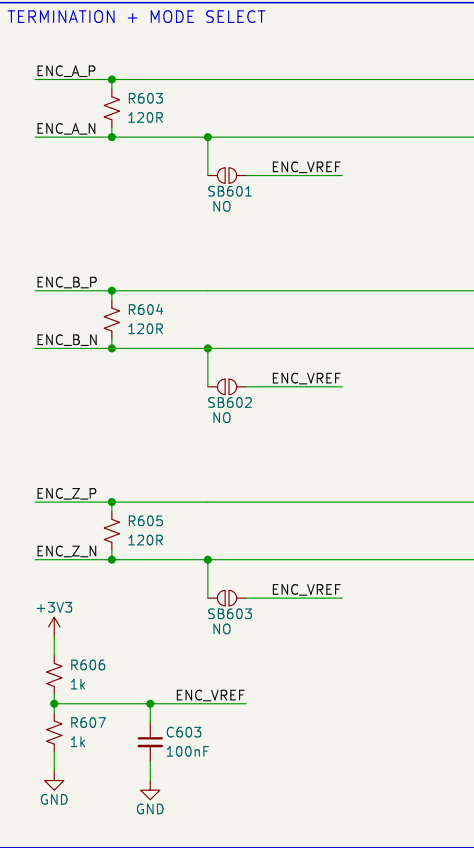
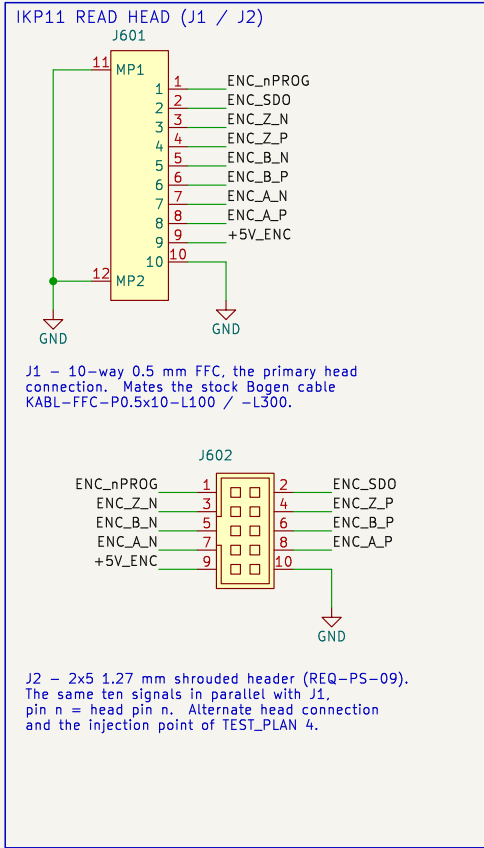
```
INTERFACES IN V24_LOGIC <- power_entry_24v (owns its PWR_FLAG).
RAIL_PG00D is open drain, the wired-AND of all four regulators
through R315 / R316 / R317 / R318. SHEET-LOCAL: LED D303, T
are its only consumers until Q0-07 allocates an MCU GPIO.
Global power nets produced here: +5V, +5VA, +3V3, +3V3SA.
GND is board-wide, one net; power_entry_24v owns its PWR_FLAG.
```

Reasoning, part sizing and residual ERC:
docs/decisions/actuators-sch-power.md

7

Size: A3	Date: 2026-09-02	Rev: 0.2
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BLOCK NOTES – linear_encoder

Ordered head: Bogen IKP11-Z1.4-P1-V5-D1-R0.5-F1000-C1 (DEC-0003, REQ-PS-05)
Z1.4 periodic index from the pole pitch, 4 counts P1 1 mm pole pitch
V5 5 V supply, < 65 mA D1 RS-422 A/B/Z
R0.5 0.5 um resolution, post-quadrature F1000 1 MHz per channel
C1 15.8 x 15.4 PCB: a 10-way 0.5 mm FFC connector AND a 1.27 mm pad row

HEAD PIN ORDER (datasheet rev 3.7 p.6 – same order on the FFC and the pads)
1 !PROG 2 SDO 3 /Z 4 Z 5 /B 6 B 7 /A 8 A 9 V+ 10 V-
Check the FFC cable contact side (type A / type D) against J1 before ordering – J1 is a bottom-contact receptacle. J2 needs no such check and is the fallback.

MODES

RS-422, the ordered D1 build (default): R603/R604/R605 fitted, SB601/SB602/SB603 open.
120 R across each pair, as the head datasheet asks (ZO = 120 R at the receiving end). No external fail-safe bias is needed: the AM26LV32 has open-circuit, shorted AND 100 R-terminated fail-safe, so an unplugged or dead head drives a static high – a stalled count, never phantom motion.
Single-ended TTL / bench injection: remove R603/R604/R605, close SB601/SB602/SB603.
Each inverting input then sits on ENC_VREF (1.65 V), so a 0–3.3 V or 0–5 V single-ended source on ENC_xP decodes correctly. This covers a TTL head variant (order code D3) and injection through J2 with no read head fitted.

U1 runs from 3V3, so its outputs are native 3V3 logic (REQ-PS-06); its –0.3 V to +5.5 V input common-mode range accepts the 5 V head directly, and it switches to 32 MHz against the 1 MHz per channel of REQ-PS-07. Both enables are tied active: G (pin 4) high, /G (pin 12) low.

No test points sit on the differential pairs: a probe stub on a terminated 1 MHz RS-422 line is a discontinuity (TEST_PLAN 2.2), and the receiver outputs carry the same information single-ended. J603, the keyed logic-analyser socket, replaces the old TP602.TP607 hooks: CH0..CH4 = LINEAR_ENCODER_A / _B / _Z, ENC_SDO, ENC_nPROG, with GND on 9 and 10 for the ground leads. TP601 stays, a 1.0 mm SMT pad for a DMM on the 5 V head supply.

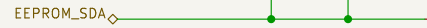
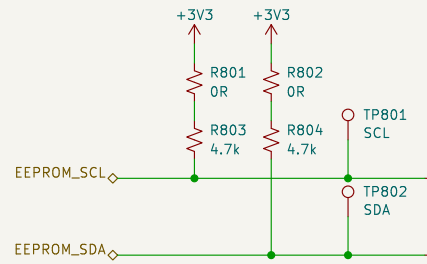
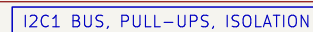
Test strategy: docs/TEST_PLAN.md
REQ-PS-01..10, REQ-AR-07, DEC-0003
RS-422 A/B/Z -> 3V3 into TIM5 (encoder mode TI12 + CH3 capture)
Bogen IKP11-Z1.4-P1-V5-D1-R0.5-F1000-C1 read head interface

Amodo Design

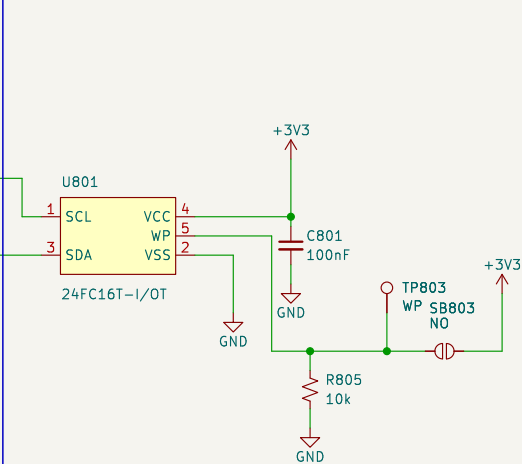
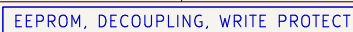
Sheet: /linear_encoder/
File: linear_encoder.kicad_sch

Title: CBs_1 – Linear Encoder (IKP11)

Size: A3 Date: 2026-09-02 Rev: 0.2
KiCad E.D.A. 9.0.8 Id: 5/10

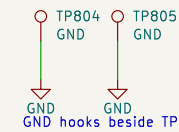


R801/R802 are the removable links of TEST_PLAN 4: lift either to drop its pull-up. SB801/SB802 are cut to take the EEPROM off I2C1 while the bus stays pulled up – this block owns that break, so mcu must not duplicate it. The SDA pull-up crosses the SCL run, no junction.



WP is ACTIVE HIGH. R805 holds it low, so the part is writable and firmware can store calibration at any time; solder SB803 to lock the device permanently. TP3 lets a jig drive WP either way. This resolves OQ-06 – no MCU pin is spent on write protect.

GROUND HOOKS



+3V3 and GND are generated in power_rails and enter this block through the hierarchy; nothing here drives them, so ERC reports power_pin_not_driven until that sheet lands. See docs/decisions/actuator-sch-periph.md.

analyse clips.

BLOCK NOTES – nvm_calibration

U1 = Microchip 24FC16T-I/OT, 16 kbit (2048 x 8) I2C EEPROM in SOT-23-5.

Pin map verified against the datasheet package drawing (DS20001703R p.1):

1 SCL 2 VSS 3 SDA 4 VCC 5 WP.
1.7-5.5 V and 1 MHz capable so it keeps

1.7–5.5 V and 1 MHz capable, so it keeps up with I2C1 at either 400 kHz or Fast-mode Plus, and 4 million erase/write cycles covers repeated calibration.

2 KB is sized for coefficients, not tables: force and temperature calibration constants, the variant (V50N / V10N – see DEC-0010, where the variants differ only by a calibration constant), board serial number and build state. Bulk force profiles live in the OCTOSP11 RAM on the mcu sheet (DEC-0006), not here.

ADDRESSING CAVEAT. The 24xx16 uses all three block-select bits internally, so it answers to the WHOLE 1010xxx address space and A0/A1/A2 do not exist on the part. No second 1010-family device can share I2C1. It is the only device on the bus today; if another is ever added, swap to a smaller 24xx02/04 first.

4k7 pull-ups suit 400 kHz on a short board-level bus with one load. They are the only pull-ups on I2C1: the mcu block must not add its own.

Test strategy: docs/TEST_PLAN.md
REQ-EL-08, REQ-AR-11; resolves OQ-06 (write protect)
I2C1: EEPROM_SCL PB8, EEPROM_SDA PB9
16 kbit I2C EEPROM for calibration and compensation data

Amodo Design

Sheet: /nvm_calibration/
File: nvm_calibration.kicad_sch

Title: CBs_1 – NVM / Calibration Store

Size: A3

Date: 2026-09-02

Rev: 0.2

KiCad E.D.A. 9.0.8

Id: 7/10

